

Chapter 21

Coulomb's Law

❑ Electric Charge

Electric charge is an intrinsic property of the fundamental particles that make up objects such as rods, silk, and fur. Therefore, the charge is a property that comes automatically with those particles wherever they exist.

Electric charge is the property of subatomic particles that causes them to experience a force when placed in an electric and magnetic field.

❑ Conductors and Insulators

➤ Conductors

Conductors are materials through which charge (free electrons in conduction band) can move freely. For example, metals (such as copper in common lamp wire), the human body, and tap water.

➤ Nonconductors/insulators

Nonconductors—also called insulators—are materials through which charge cannot move freely (**does not have any free electrons or conduction electrons**). For example, rubber (such as the insulation on common lamp wire), plastic, glass, and chemically pure water.

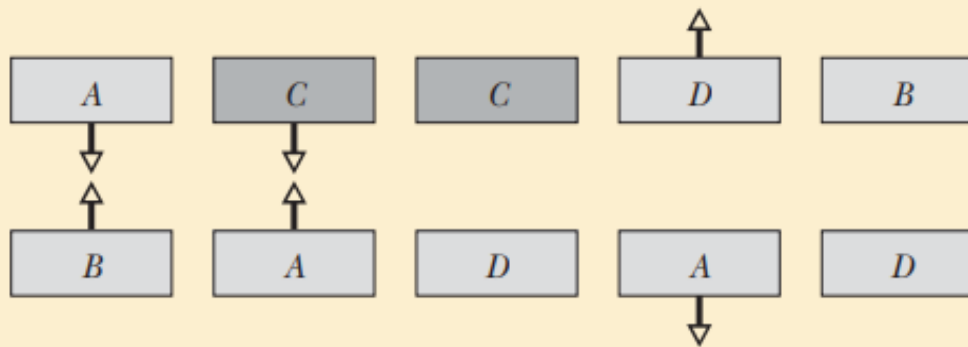
Note:

Semiconductors are materials that are intermediate between conductors and insulators; examples include silicon and germanium in computer chips. Superconductors are materials that are perfect conductors, allowing charge to move without any hindrance. In these chapters, we discuss only conductors and insulators.



Checkpoint 1

The figure shows five pairs of plates: *A*, *B*, and *D* are charged plastic plates and *C* is an electrically neutral copper plate. The electrostatic forces between the pairs of plates are shown for three of the pairs. For the remaining two pairs, do the plates repel or attract each other?



□Coulomb's Law

Let us consider the magnitude of two charges are q_1 and q_2 as shown in **Figure 1**, where the distance between them is, r .

Coulomb's law states that the magnitude of the electric force (attractive or repulsive) between two point charges is proportional to the multiplication of the magnitude of two charges and inversely proportional to the square of the distance between those charges.

Electrostatic constant,

Permittivity constant,

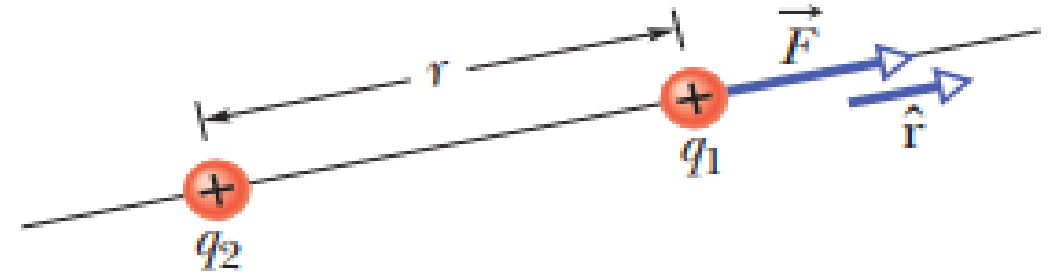


Figure 1: The electrostatic force on particle 1 can be described in terms of a unit vector along an axis through the two particles, radially away from particle 2.

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2.$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2.$$

Vector format of force :

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$$

Multiple Forces. The electrostatic force obeys the principle of superposition. Suppose we have n charged particles near a chosen particle called particle 1; then the net force on particle 1 is given by the vector sum

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{15} + \cdots + \vec{F}_{1n},$$

❖ Shell Theories for *electrostatic force*

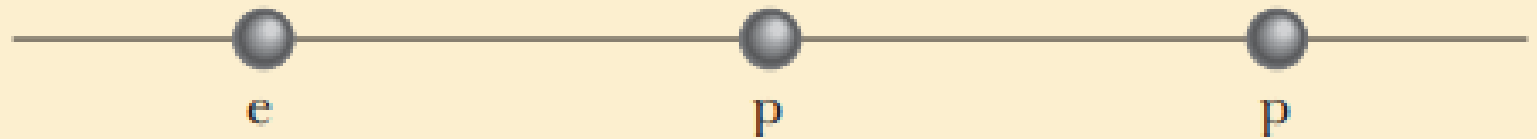
Shell theory 1. A charged particle outside a shell with charge uniformly distributed on its surface is attracted or repelled as if the shell's charge were concentrated as a particle at its center.

Shell theory 2. A charged particle inside a shell with charge uniformly distributed on its surface has no net force acting on it due to the shell.



Checkpoint 2

The figure shows two protons (symbol p) and one electron (symbol e) on an axis.



On the central proton, what is the direction of (a) the force due to the electron, (b) the force due to the other proton, and (c) the net force?

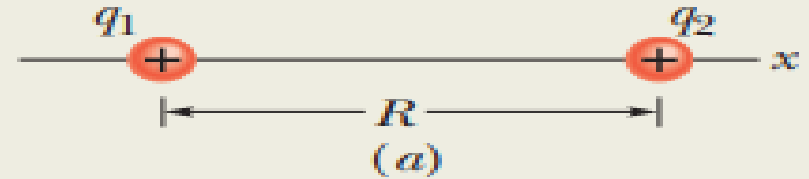
Finding the net force due to two other particles

Sample Problem 21.01

This sample problem actually contains three examples, to build from basic stuff to harder stuff. In each we have the same charged particle 1. First there is a single force acting on it (easy stuff). Then there are two forces, but they are just in opposite directions (not too bad). Then there are again two forces but they are in very different directions (ah, now we have to get serious about the fact that they are vectors). The key to all three examples is to draw the forces correctly *before* you reach for a calculator, otherwise you may be calculating nonsense on the calculator. (Figure 21-7 is available in *WileyPLUS* as an animation with voiceover.)

(a) Figure 21-7a shows two positively charged particles fixed in place on an x axis. The charges are $q_1 = 1.60 \times 10^{-19} \text{ C}$ and $q_2 = 3.20 \times 10^{-19} \text{ C}$, and the particle separation is $R = 0.0200 \text{ m}$. What are the magnitude and direction of the electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2?

This is the first arrangement.



$$\begin{aligned} F_{12} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{R^2} \\ &= (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \\ &\quad \times \frac{(1.60 \times 10^{-19} \text{ C})(3.20 \times 10^{-19} \text{ C})}{(0.0200 \text{ m})^2} \\ &= 1.15 \times 10^{-24} \text{ N}. \end{aligned}$$

Thus, force $\vec{F}_{12}$ has the following magnitude and direction (relative to the positive direction of the x axis):

$$1.15 \times 10^{-24} \text{ N} \quad \text{and} \quad 180^\circ. \quad (\text{Answer})$$

We can also write $\vec{F}_{12}$ in unit-vector notation as

$$\vec{F}_{12} = -(1.15 \times 10^{-24} \text{ N})\hat{i}. \quad (\text{Answer})$$

(b) Figure 21-7c is identical to Fig. 21-7a except that particle 3 now lies on the x axis between particles 1 and 2. Particle 3 has charge $q_3 = -3.20 \times 10^{-19} \text{ C}$ and is at a distance $\frac{3}{4}R$ from particle 1. What is the net electrostatic force $\vec{F}_{1,\text{net}}$ on particle 1 due to particles 2 and 3?

$$\begin{aligned} F_{13} &= \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_3|}{(\frac{3}{4}R)^2} \\ &= (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \\ &\quad \times \frac{(1.60 \times 10^{-19} \text{ C})(3.20 \times 10^{-19} \text{ C})}{(\frac{3}{4})^2(0.0200 \text{ m})^2} \\ &= 2.05 \times 10^{-24} \text{ N}. \end{aligned}$$

We can also write $\vec{F}_{13}$ in unit-vector notation:

$$\vec{F}_{13} = (2.05 \times 10^{-24} \text{ N})\hat{i}.$$

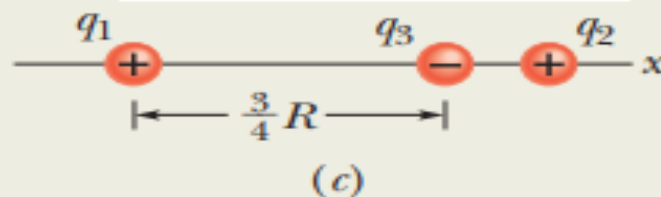
The net force $\vec{F}_{1,\text{net}}$ on particle 1 is the vector sum of $\vec{F}_{12}$ and $\vec{F}_{13}$; that is, from Eq. 21-7, we can write the net force $\vec{F}_{1,\text{net}}$ on particle 1 in unit-vector notation as

$$\begin{aligned} \vec{F}_{1,\text{net}} &= \vec{F}_{12} + \vec{F}_{13} \\ &= -(1.15 \times 10^{-24} \text{ N})\hat{i} + (2.05 \times 10^{-24} \text{ N})\hat{i} \\ &= (9.00 \times 10^{-25} \text{ N})\hat{i}. \end{aligned} \quad (\text{Answer})$$

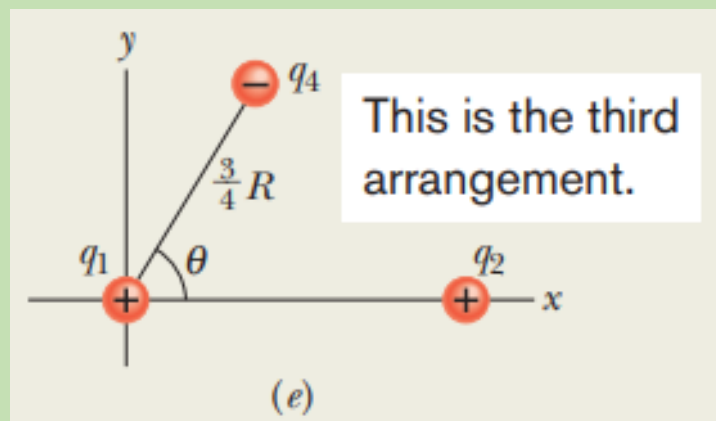
Thus, $\vec{F}_{1,\text{net}}$ has the following magnitude and direction (relative to the positive direction of the x axis):

$$9.00 \times 10^{-25} \text{ N} \quad \text{and} \quad 0^\circ. \quad (\text{Answer})$$

This is the second arrangement.



(c) Figure 21-7e is identical to Fig. 21-7a except that particle 4 is now included. It has charge $q_4 = -3.20 \times 10^{-19} \text{ C}$, is at a distance $\frac{3}{4}R$ from particle 1, and lies on a line that makes an angle $\theta = 60^\circ$ with the x axis. What is the net electrostatic force $\vec{F}_{1,\text{net}}$ on particle 1 due to particles 2 and 4?



This is the third arrangement.

$$\vec{F}_{1,\text{net}} = \vec{F}_{12} + \vec{F}_{14}.$$

Method 1. *Summing directly on a vector-capable calculator.* For $\vec{F}_{12}$, we enter the magnitude 1.15×10^{-24} and the angle 180° . For $\vec{F}_{14}$, we enter the magnitude 2.05×10^{-24} and the angle 60° . Then we add the vectors.

Method 2. *Summing in unit-vector notation.* First we rewrite $\vec{F}_{14}$ as

$$\vec{F}_{14} = (F_{14} \cos \theta)\hat{i} + (F_{14} \sin \theta)\hat{j}.$$

Substituting $2.05 \times 10^{-24} \text{ N}$ for F_{14} and 60° for θ , this becomes

$$\vec{F}_{14} = (1.025 \times 10^{-24} \text{ N})\hat{i} + (1.775 \times 10^{-24} \text{ N})\hat{j}.$$

Then we sum:

$$\begin{aligned}\vec{F}_{1,\text{net}} &= \vec{F}_{12} + \vec{F}_{14} \\ &= -(1.15 \times 10^{-24} \text{ N})\hat{i} \\ &\quad + (1.025 \times 10^{-24} \text{ N})\hat{i} + (1.775 \times 10^{-24} \text{ N})\hat{j} \\ &\approx (-1.25 \times 10^{-25} \text{ N})\hat{i} + (1.78 \times 10^{-24} \text{ N})\hat{j}. \\ &\hspace{15em} \text{(Answer)}\end{aligned}$$

Method 3. *Summing components axis by axis.* The sum of the x components gives us

$$\begin{aligned}F_{1,\text{net},x} &= F_{12,x} + F_{14,x} = F_{12} + F_{14} \cos 60^\circ \\ &= -1.15 \times 10^{-24} \text{ N} + (2.05 \times 10^{-24} \text{ N})(\cos 60^\circ) \\ &= -1.25 \times 10^{-25} \text{ N}.\end{aligned}$$

The sum of the y components gives us

$$\begin{aligned}F_{1,\text{net},y} &= F_{12,y} + F_{14,y} = 0 + F_{14} \sin 60^\circ \\ &= (2.05 \times 10^{-24} \text{ N})(\sin 60^\circ) \\ &= 1.78 \times 10^{-24} \text{ N}.\end{aligned}$$

The net force $\vec{F}_{1,\text{net}}$ has the magnitude

$$F_{1,\text{net}} = \sqrt{F_{1,\text{net},x}^2 + F_{1,\text{net},y}^2} = 1.78 \times 10^{-24} \text{ N}. \quad \text{(Answer)}$$

To find the direction of $\vec{F}_{1,\text{net}}$, we take

$$\theta = \tan^{-1} \frac{F_{1,\text{net},y}}{F_{1,\text{net},x}} = -86.0^\circ.$$

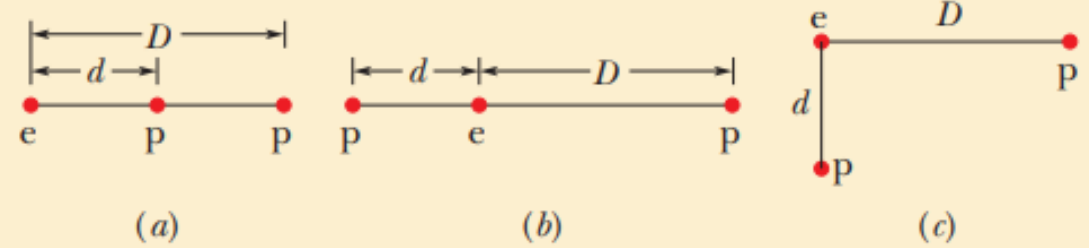
However, this is an unreasonable result because $\vec{F}_{1,\text{net}}$ must have a direction between the directions of $\vec{F}_{12}$ and $\vec{F}_{14}$. To correct θ , we add 180° , obtaining

$$-86.0^\circ + 180^\circ = 94.0^\circ. \quad \text{(Answer)}$$



Checkpoint 3

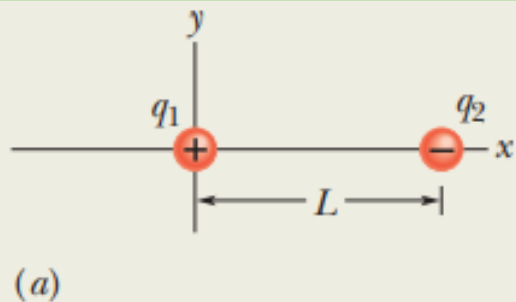
The figure here shows three arrangements of an electron e and two protons p . (a) Rank the arrangements according to the magnitude of the net electrostatic force on the electron due to the protons, largest first. (b) In situation c , is the angle between the net force on the electron and the line labeled d less than or more than 45° ?



Equilibrium of two forces on a particle

Sample Problem 21.02

Figure 21-8a shows two particles fixed in place: a particle of charge $q_1 = +8q$ at the origin and a particle of charge $q_2 = -2q$ at $x = L$. At what point (other than infinitely far away) can a proton be placed so that it is in *equilibrium* (the net force on it is zero)? Is that equilibrium *stable* or *unstable*? (That is, if the proton is displaced, do the forces drive it back to the point of equilibrium or drive it farther away?)



$$\left(\frac{x - L}{x}\right)^2 = \frac{1}{4}.$$

$$\frac{x - L}{x} = \frac{1}{2}$$

and

$$x = 2L. \quad (\text{Answer})$$

$$\frac{1}{4\pi\epsilon_0} \frac{8qq_p}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{2qq_p}{(x - L)^2}.$$

The equilibrium at $x = 2L$ is *unstable*; that is, if the proton is displaced leftward from point R , then F_1 and F_2 both increase but F_2 increases more (because q_2 is closer than q_1), and a net force will drive the proton farther leftward. If the proton is displaced rightward, both F_1 and F_2 decrease but F_2 decreases more, and a net force will then drive the proton farther rightward. In a *stable* equilibrium, if the proton is displaced slightly, it returns to the equilibrium position.

Charge sharing by two identical conducting spheres

Sample Problem 21.03

In Fig. 21-9*a*, two identical, electrically isolated conducting spheres *A* and *B* are separated by a (center-to-center) distance *a* that is large compared to the spheres. Sphere *A* has a positive charge of $+Q$, and sphere *B* is electrically neutral. Initially, there is no electrostatic force between the spheres. (The large separation means there is no induced charge.)

(a) Suppose the spheres are connected for a moment by a conducting wire. The wire is thin enough so that any net charge on it is negligible. What is the electrostatic force between the spheres after the wire is removed?

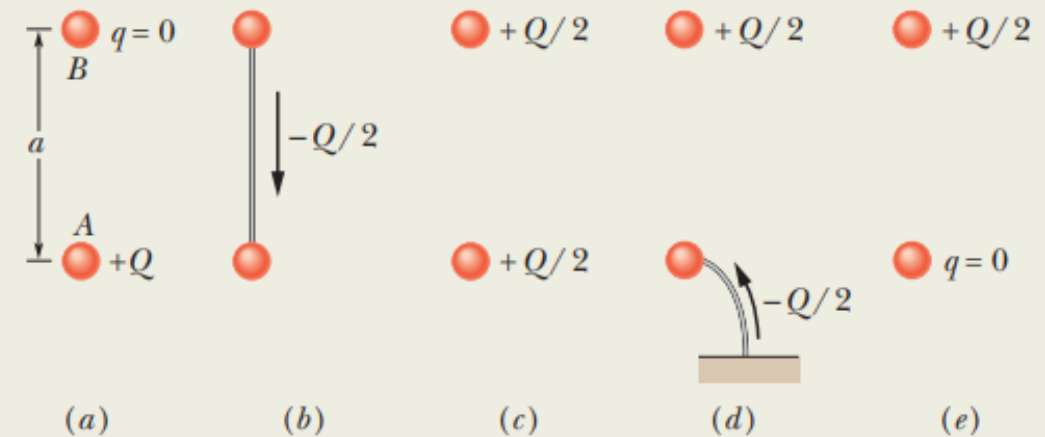


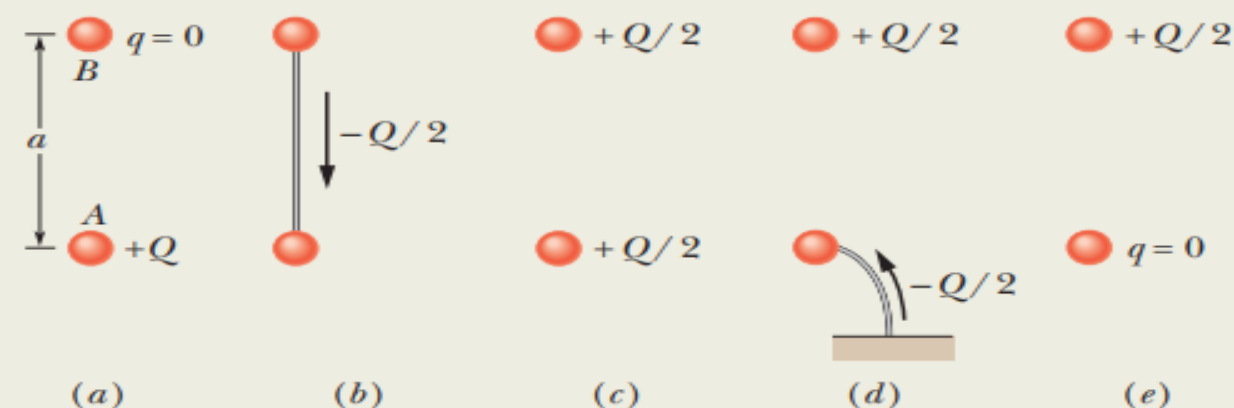
Figure 21-9 Two small conducting spheres *A* and *B*. (a) To start, sphere *A* is charged positively. (b) Negative charge is transferred from *B* to *A* through a connecting wire. (c) Both spheres are then charged positively. (d) Negative charge is transferred through a grounding wire to sphere *A*. (e) Sphere *A* is then neutral.

KEY IDEAS

(1) Because the spheres are identical, connecting them means that they end up with identical charges (same sign and same amount). (2) The initial sum of the charges (including the signs of the charges) must equal the final sum of the charges.

Reasoning: When the spheres are wired together, the (negative) conduction electrons on B , which repel one another, have a way to move away from one another (along the wire to positively charged A , which attracts them—Fig. 21-9b). As B loses negative charge, it becomes positively charged, and as A gains negative charge, it becomes *less* positively charged. The transfer of charge stops when the charge on B has increased to $+Q/2$ and the charge on A has decreased to $+Q/2$, which occurs when $-Q/2$ has shifted from B to A .

After the wire has been removed (Fig. 21-9c), we can assume that the charge on either sphere does not disturb the uniformity of the charge distribution on the other sphere, because the spheres are small relative to their separation. Thus, we can apply the first shell theorem to each sphere. By Eq. 21-4 with $q_1 = q_2 = Q/2$ and $r = a$,



$$F = \frac{1}{4\pi\epsilon_0} \frac{(Q/2)(Q/2)}{a^2} = \frac{1}{16\pi\epsilon_0} \left(\frac{Q}{a}\right)^2. \quad (\text{Answer})$$

The spheres, now positively charged, repel each other.

(b) Next, suppose sphere A is grounded momentarily, and then the ground connection is removed. What now is the electrostatic force between the spheres?

Reasoning: When we provide a conducting path between a charged object and the ground (which is a huge conductor), we neutralize the object. Were sphere A negatively charged, the mutual repulsion between the excess electrons would cause them to move from the sphere to the ground. However, because sphere A is positively charged, electrons with a total charge of $-Q/2$ move *from* the ground up onto the sphere (Fig. 21-9d), leaving the sphere with a charge of 0 (Fig. 21-9e). Thus, the electrostatic force is again zero.

□ Charge Is Quantized

Any positive or negative charge that can be detected can be written as

$$q = ne, \quad n = \pm 1, \pm 2, \pm 3, \dots \dots \dots$$

In which e , the **elementary charge**, has the approximate value

$$e = 1.602 \times 10^{-19} \text{ C}.$$

Note:

You should, however, keep in mind what is intended: Particles are the substance and charge happens to be one of their properties, just as mass is.

When a physical quantity such as charge can have only discrete values rather than any value, we say that the quantity is quantized. It is possible, for example, to find a particle that has no charge at all or a charge of $+10e$ or $-6e$, but not a particle with a charge of, say, $3.57e$.



Checkpoint 4

Initially, sphere A has a charge of $-50e$ and sphere B has a charge of $+20e$. The spheres are made of conducting material and are identical in size. If the spheres then touch, what is the resulting charge on sphere A ?

Mutual electric repulsion in a nucleus

Sample Problem 21.04

The nucleus in an iron atom has a radius of about $4.0 \times 10^{-15} \text{ m}$ and contains 26 protons.

(a) What is the magnitude of the repulsive electrostatic force between two of the protons that are separated by $4.0 \times 10^{-15} \text{ m}$?

$$\begin{aligned} F &= \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \\ &= \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.602 \times 10^{-19} \text{ C})^2}{(4.0 \times 10^{-15} \text{ m})^2} \\ &= 14 \text{ N.} \end{aligned} \quad (\text{Answer})$$

(b) What is the magnitude of the gravitational force between those same two protons?

$$\begin{aligned} F &= G \frac{m_p^2}{r^2} \\ &= \frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(1.67 \times 10^{-27} \text{ kg})^2}{(4.0 \times 10^{-15} \text{ m})^2} \\ &= 1.2 \times 10^{-35} \text{ N.} \end{aligned} \quad (\text{Answer})$$

□ Charge Is Conserved

A charge is a property associated with the matter due to which it produces and experiences electrical and magnetic effects. The basic idea behind the conservation of charge is that the total charge of the system is conserved. We can define it as:

Conservation of charge is the principle that the total electric charge in an isolated system never changes. The net quantity of electric charge, the amount of positive charge minus the amount of negative charge in the universe, is always conserved.

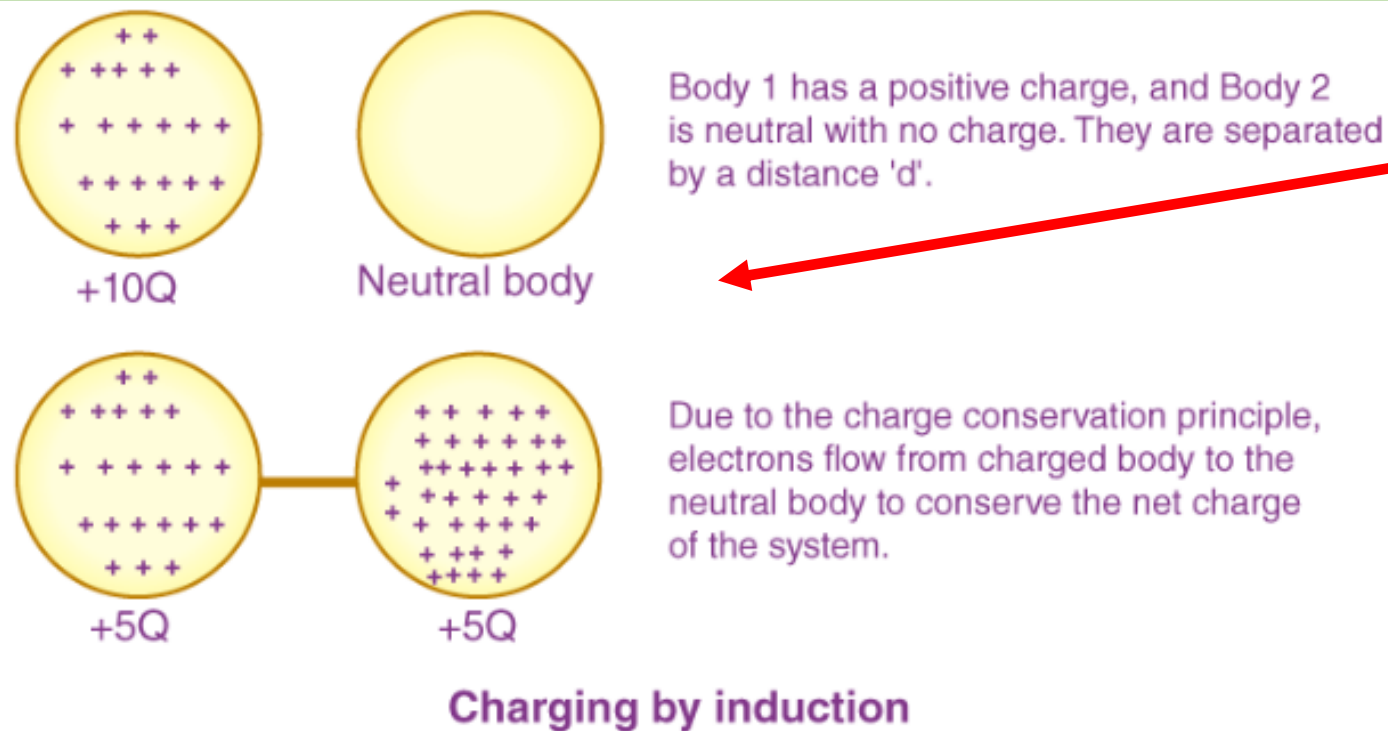
The law of conservation of charge says that the net charge of an isolated system will always remain constant. Let's try to understand it in more depth. There is a possibility of two ideal states for a system for multiple objects.

- The first one is that all the object has a net neutral charge. So in the whole system, there is the same number of protons and electrons. So for each proton, there is an electron to balance it.
- Another ideal state would be the net charge of the system being distributed uniformly in the objects. So rather than concentrating negative charge in a few bodies, the charge on the body is evenly distributed throughout by the transfer of the electron, and this can be achieved by the transfer of electrons from higher to lower polarity. Only electrons can be involved in the transfer charges, not protons.

❖ Examples of the conservation of charge

Conservation of charge means one can't produce a net charge. Some of its examples are provided below.

- Charges due to induction
- During radioactive decay, a proton decays into a positron and neutron, but no net charge production.



From the left side image, we can say that if our system is not in the influence of any other charges, the net internal distribution among charge will go on in such a way that the entire net charge of the system will remain the same. In other words, we can say that: ***Charge can neither be created nor be destroyed and there is total conservation of charge.***

Thank You